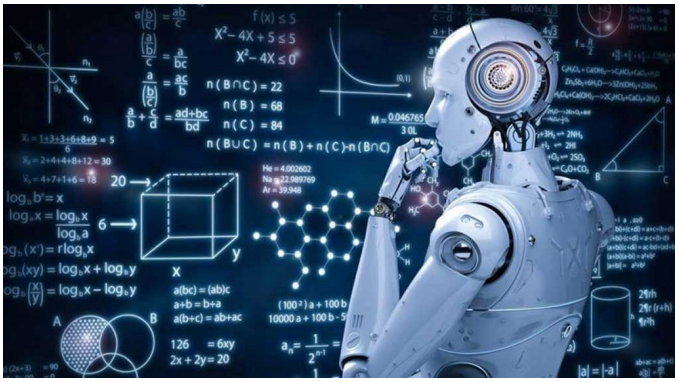
UNIT 4

ROBOTICS



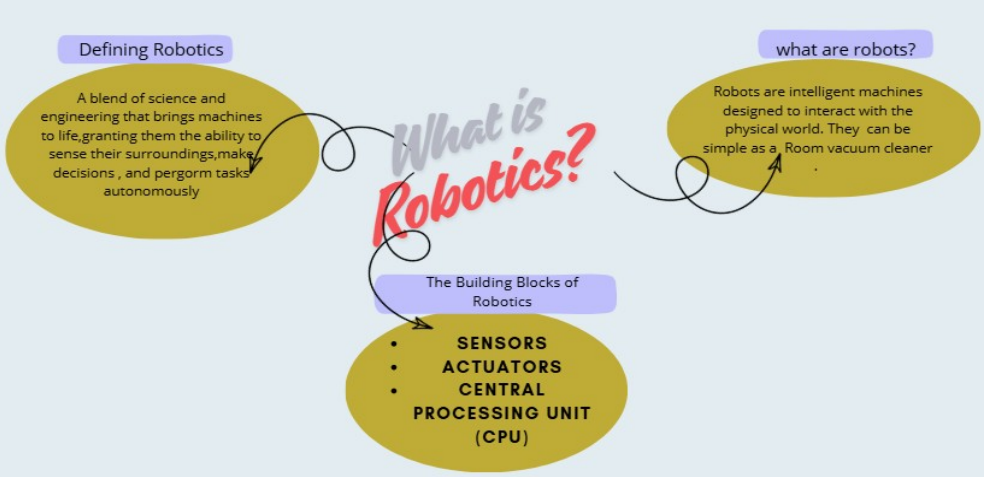
GLOSSARY

WORD	CONCEPT
ACCELEROMETER	A device for measuring acceleration or force. These are related by Newton's second law: force = mass * acceleration.
ACCURACY	The precision with which a computed or calculated robot position can be attained.
ACTUATOR	A motor that reads programming signals and translates them into mechanical movement.
ANDROID	A humanoid robot designed to resemble an adult human male.
CONTROLLER SYSTEM	A computer of some type that stores data, executes programs, and directs the operations of the robot.
CYBORG	Shorthand for 'cybernetic organism', it is any being that possesses both biological and artificial parts.
COBOTS	robots that interface directly with humans.
END EFFECTOR	An end effector is a device at the end of a robotic arm, designed to interact with the environment, such as our patented suction picker.
CONTROLLER	The main device that processes information and carries out instructions in a robot.
CONTROL SYSTEM	A method of directing the type of path a robot takes.
JOINT	The location at which two or more parts of a robotic arm make contact. Joints allow parts to move in different directions.
KINEMATICS	In robotics, kinematics involves studying the mapping of coordinates in motion.
SIMULATOR	A software application that creates a virtual world in which robots can be tested.
ROBOTICS	Science of designing, building and applying robots.
GRIPPER	Gripper (usually with two fingers) grasping objects of different shape, mass and material. It is actuated by either pneumatic, hydraulic or electrical motors. It can be equipped with sensors of force or of proximity
ACTIVE SENSOR	A sensor which instigates an action and then waits for a response such as transmitting a signal and measuring the response when it comes back.

ACCELERATION	A change in velocity .
BRAIN-COMPUTER INTERFACE	A direct connection between a brain and a computer, allowing the brain to command the computer or the computer to monitor the brain.
SENSOR	A device which is used to measure a quantity – such as the distance to an object or the speed of a robot.

*if you want to know about the history of robotics visit
https://www.g2.com/articles/history-of-robots?hsCtaTracking=314fbdf4- ec3d-40a6-bbdb-5141a44d5781%7C0de54460-a6b2-49cf-9256-4580d2a73c0f

4.1. What is Robotics?



Robotics is an interdisciplinary sector of science and engineering dedicated to the design, construction and use of mechanical robots. Our guide will give you a concrete grasp of robotics, including different types of robots and how they're being applied across industries.

Robotics is the intersection of science, engineering and technology that produces machines, called robots, that replicate or substitute for human actions. Robots perform basic and repetitive tasks with greater efficiency and accuracy than humans, making them ideal for industries like manufacturing. However, the introduction of artificial intelligence in robotics has given robots the ability to handle increasingly complex situations in various industries. (October,2024)

Robotics is the engineering and operation of machines that can autonomously or semi-autonomously perform physical tasks on behalf of a human. Typically, robots perform tasks that are either highly repetitive or too dangerous for a human to carry out safely. The study of Robots is called Robotics.

Robotics is an interdisciplinary branch of engineering and science that includes mechanical engineering,

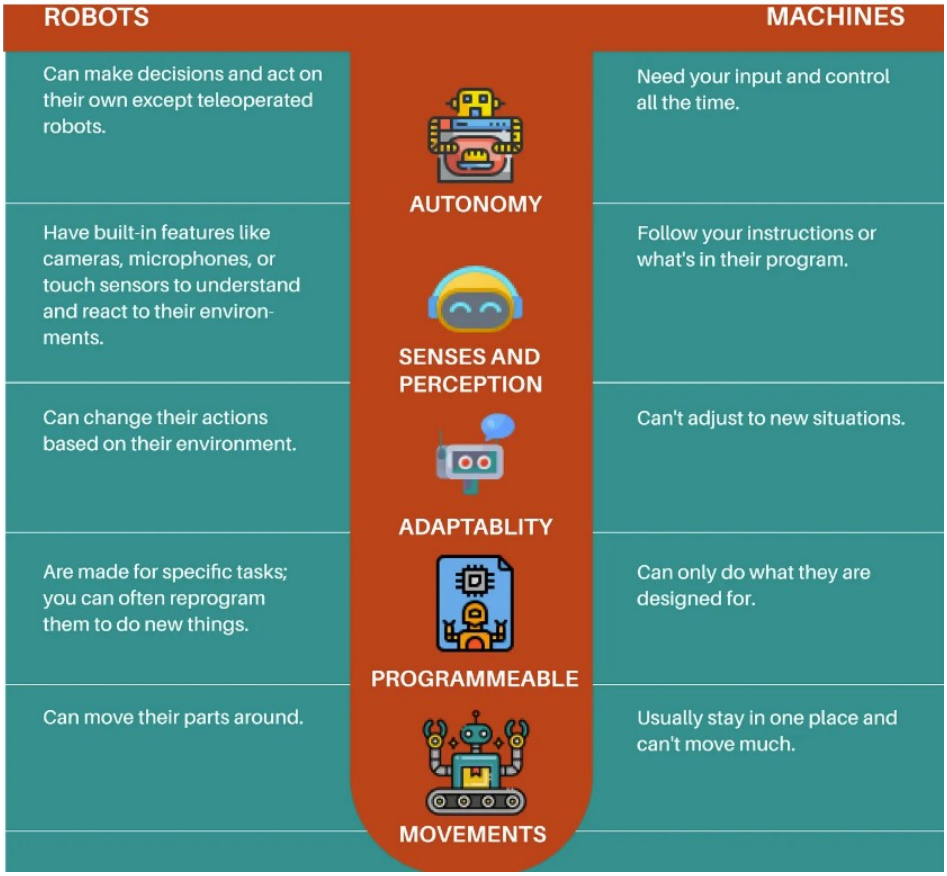
electronic engineering, information engineering, computer science, and others.

Robotics is the engineering and operation of machines that can autonomously or semi- autonomously perform physical tasks on behalf of a human. Typically, robots perform tasks that are either highly repetitive or too dangerous for a human to carry out safely.

The field of robotics has greatly advanced with several new general technological achievements. One is the rise of big data, which offers more opportunity to build programming capability into robotic systems. Another is the use of new kinds of sensors and connected devices to monitor environmental aspects like temperature, air pressure, light, motion and more. All of this serves robotics and the generation of more complex and sophisticated robots for many uses, including manufacturing, health and saf Robots' Anatomy and Components ety, and human assistance.

4.1.1. The Difference Between a Robot and a Machine.

All robots are machines, but not all machines are robots, with the primary difference being that robots possess the autonomy and sensory feedback to sense, compute, and act independently.



4.2. Function of robotics

Robotics is an interdisciplinary research area at the interface of computer science and engineering. **It** involves design, construction, operation, and use of **robots**. The goal of **robotics** is to design intelligent machines that can help and assist humans in their day-to-day lives and keep everyone safe.

Robot is a machine built to carry out a complex task (or set of tasks) by physically moving and interacting with the world around it. Robots can usually be programmed by a user.

Robot is automatically operated machine that replaces human effort, though it may not resemble human beings in appearance or perform functions in a humanlike manner. By extension, robotics is the engineering discipline dealing with the design, construction, and operation of robots.

4.3. Robots' Anatomy and Components

To understand robotics better, it's essential to grasp the building blocks that make these machines tick:

Sensors: Imagine if robots were blind, deaf, and unable to sense the world around them. Sensors are the eyes and ears of robots. They come in various forms, such as cameras, ultrasonic sensors, and touch sensors, allowing robots to gather data about their environment.

Actuators: Think of actuators as the muscles of a robot. These components enable robots to move, manipulate objects, and perform tasks. Electric motors, hydraulic systems, and pneumatic actuators are all examples of these critical components.

Central Processing Unit (CPU): Every robot needs a brain, and that's where the CPU comes into play. This is where the magic happens. It's where data from sensors is processed, decisions are made, and actions are executed.

4.4. Common types of robots

The most common types of robots include:

4.4.1.Autonomous Mobile Robots (AMRs)

Using sophisticated sensors, artificial intelligence, and machine learning, autonomous mobile robots (AMRs) can support many tasks, including delivering items, carrying out safety and security checks, inventory automation, rescue missions, crop harvesting, and even space exploration.

4.4.2. Automated Guided Vehicles (AGVs)

As a form of autonomous mobile robots, automated guided vehicles (AGVs) are movable robots capable of

navigating mapped routes via wires on a floor, radio waves, cameras, magnets, or lasers. These robots have several applications, including transporting heavy materials in factories and warehouses.

4.4.3. Articulated Robots

One of the main features of an articulated robot is rotary joints that can range from simple two-jointed structures to complex systems of multiple interacting joints and materials. The articulated robot is a typical type of industrial robot and often resembles a human arm.

4.4.4. Humanoids

A humanoid robot resembles a generic human body form. The specialized design is typically used for mimicking human motions and interactions. A common use case of humanoid robots is automating tasks to increase efficiency and cost-savings.

4.4.5. Cobots

A cobot is a contracted term for a collaborative robot. These robots work side-by-side with human workers. They differ from other robots because their function is not intended to do work in place of human employees. Often, they are compact and perform a variety of tasks in the metal industry, automotive, electronics, laboratories, and hospitals.

4.5. What is Robotics Engineering?

Robotics engineering is a multidisciplinary field including electrical, mechanical, and computer engineering. It deals with designing, building, operating, and engineering robots and robotic systems based on theoretical understanding and practical application.

4.5.1. What do Robotics Engineers do?

A robotics engineer develops robotic applications across many industries, including automotive, aerospace, manufacturing, defense, agriculture, and healthcare. Robotics engineers work on designing, building, and operating robots and robotic systems.

Robotic engineers develop robots and robotic systems using a combination of mechanical, electrical, and computer engineering principles and technologies including selection and integration of the necessary components, such as sensors, actuators, motors, and controllers.

4.5. Future of Robotics

https://www.youtube.com/watch?v=EWX0KbLMc58

The Text: The Convergence of Autonomy and Systems Architecture

The discipline of robotics has evolved from the pursuit of isolated mechanical automation to a sophisticated field deeply embedded within the broader scope of systems engineering. Historically, a robot was viewed merely as a programmable manipulator—a discrete component

designed to execute repetitive tasks within a controlled environment. However, contemporary systems engineering demands a holistic perspective where the robot is not an isolated entity but a node within a complex, interconnected ecosystem. This paradigm shift is largely driven by the proliferation of autonomous systems, which require seamless interaction between mechanical hardware, sensory perception, control algorithms, and human operators.

Central to this evolution is the concept of **Cyber-Physical Systems (CPS)** . In a CPS framework, computational elements collaborate to monitor and control physical processes, often with feedback loops where physical actions affect computations and vice versa. For the systems engineer, the robot becomes a tangible manifestation of CPS principles. The challenge lies not only in optimizing the robot's individual performance but in ensuring its interoperability with other subsystems, such as enterprise resource planning (ERP) software, safety protocols, and dynamic environmental variables.

One of the most pressing issues in modern robotics is **uncertainty management**. Unlike traditional software engineering, where inputs are often deterministic, robotic systems operate in unstructured, dynamic environments. A warehouse robot, for instance, must navigate around unpredictable human workers, adapt to shifting floor layouts, and handle objects of varying weights and textures. From a systems engineering perspective, this requires a shift from rigid, rule-based programming to probabilistic models and machine learning algorithms that can make decisions under uncertainty. The engineer must therefore design architectures that are robust, fault-tolerant, and capable of graceful degradation—meaning that if one component fails, the system does not collapse entirely but continues to operate at a reduced capacity.

Furthermore, the integration of robotics into larger systems raises significant **ethical and regulatory considerations**. As robots become more autonomous, the question of accountability becomes paramount. If an autonomous drone makes a decision that results in property damage or injury, who is responsible? The manufacturer, the software developer, the operator, or the robot itself? Systems engineers are increasingly required to incorporate ethical frameworks into their design processes, ensuring that autonomous behaviors align with legal standards and societal norms. This is not merely a philosophical exercise; it is a practical engineering requirement that influences everything from sensor fusion to decision-making hierarchies.

Finally, the future of robotics in systems engineering is inextricably linked to **modularity and scalability**. The traditional monolithic approach—where a robot is designed for a single, specific task—is giving way to modular architectures where hardware and software components can be reconfigured for different applications. This aligns with the systems engineering principle of separation of concerns, allowing for easier maintenance, upgrades, and customization. As the Internet of Things (IoT) continues to mature, robots will increasingly function as part of a larger network of smart devices, sharing data and coordinating actions in real time. For the systems engineer, this represents both an unprecedented opportunity and a formidable challenge: to design systems that are not only intelligent and adaptable but also secure, ethical, and resilient.

Instructions: Answer the following questions in complete sentences. Use your own words as far as possible. Each answer should demonstrate critical engagement with the text.

1. According to the first paragraph, what fundamental transformation has occurred in the way systems engineers perceive robots within modern engineering frameworks?

2. Explain the concept of Cyber-Physical Systems (CPS) as described in the text. Why does the author consider the robot a "tangible manifestation" of CPS principles?

3. What specific challenges does a warehouse robot face that illustrate the concept of "uncertainty management"? How do these challenges differ from traditional software engineering scenarios?

4. The author states that systems engineers must design architectures capable of "graceful degradation." What does this term mean in the context of robotics, and why is it considered essential?

5. Why does the increasing autonomy of robots raise questions of accountability? Provide an example from the text that illustrates this ethical dilemma.

6. According to the text, why is incorporating ethical frameworks into design processes "not merely a philosophical exercise"? What practical engineering implications does this have?

7. How does the principle of "separation of concerns" relate to the shift from monolithic robot designs to modular architectures? What advantages does this shift offer?

8. In what ways does the maturation of the Internet of Things (IoT) influence the future of robotics in systems engineering? What opportunities and challenges does this present?

9. The author describes the systems engineer's challenge as designing systems that are "not only intelligent and adaptable but also secure, ethical, and resilient." Why do you think the author groups these five attributes together? Are they equally important?

10. Based on the entire text, what do you consider to be the most significant paradigm shift in robotics from a system engineering perspective? Justify your answer with evidence from at least two paragraphs.

Part B: Fill in the Gaps

Instructions: Complete each sentence with the most appropriate word or phrase from the box below. Each item may require one or more words. Use each option ONLY ONCE.

Word Bank

Deterministic Interoperability probabilistic fault-tolerant graceful degradation monolithic

separation of concerns sensor fusion feedback loops modularity

1. In traditional software engineering, inputs are often _____, meaning they are predictable and follow fixed rules, unlike the unpredictable inputs encountered in robotic systems.

2. The challenge for systems engineers lies not only in optimizing individual robot performance but also in ensuring its _____ with other subsystems such as ERP software and safety protocols.

3. To manage uncertainty in unstructured environments, engineers must shift from rigid

Word Bank

rule-based programming to _____ models and machine learning algorithms.

4. A _____ system is one that continues to operate at a reduced capacity rather than collapsing entirely when one component fails.

5. The concept of _____ refers to the ability of a system to maintain functionality and recover from failures without complete shutdown.

6. In a Cyber-Physical System, _____ allow physical actions to affect computations and vice versa, creating a continuous cycle of monitoring and control.

7. The traditional _____ approach, where a robot is designed for a single specific task, is giving way to more flexible architectures.

8. The systems engineering principle of _____ allows for easier maintenance, upgrades, and customization by breaking down systems into distinct, independent modules.

9. _____ is the process of combining data from multiple sensors to create a more accurate and reliable perception of the environment.

10. As robots become part of larger networks of smart devices, _____ becomes essential for allowing hardware and software components to be reconfigured for different applications.

Answer Key

Part A: Open Questions — Model Answers

1.
Historically, robots were regarded as isolated, programmable manipulators designed for repetitive tasks in controlled environments. Contemporary systems engineering, however, views robots as interconnected nodes within a complex ecosystem, requiring seamless interaction between hardware, software, sensors, and human operators.

2.
Cyber-Physical Systems (CPS) are frameworks in which computational elements collaborate to monitor and control physical processes, often through feedback loops. The author considers the robot a "tangible manifestation" of CPS because it embodies the integration of digital computation with physical action, demonstrating how computational decisions directly influence real-world operations.

3.
A warehouse robot must navigate around unpredictable human workers, adapt to shifting floor layouts, and handle objects of varying weights and textures. These challenges differ from traditional software engineering because traditional inputs are often deterministic, whereas robotic systems operate in unstructured, dynamic environments requiring probabilistic decision-making.

4.
"Graceful degradation" refers to a system's ability to continue operating at a reduced capacity when one component fails, rather than collapsing entirely. It is essential because robotic systems operate in unpredictable environments where failures can occur unexpectedly, and complete shutdown could be dangerous or costly.

5.
Increasing autonomy raises questions of accountability because when an autonomous robot makes a decision resulting in harm, it becomes unclear who is responsible—the manufacturer, software developer, operator, or the robot itself. The text provides the example of an autonomous drone causing property damage or injury.

6.
Incorporating ethical frameworks is not merely philosophical because it directly influences practical engineering decisions. For instance, ethical requirements shape sensor fusion design (ensuring sensors reliably detect humans) and decision-making hierarchies (prioritizing human safety over task completion), making ethics an integral part of the technical design process.

7.
"Separation of concerns" relates to modular architectures by advocating for breaking down systems into distinct, independent modules rather than a single monolithic block. This shift offers advantages such as easier maintenance, upgrades, customization, and scalability, as changing one module does not require redesigning the entire system.

8.
The maturation of IoT influences robotics by enabling robots to function as part of larger networks of smart devices, sharing data and coordinating actions in real time. This presents opportunities for enhanced collaboration and efficiency, but also challenges related to security, ethical standards, and system resilience.

9.
The author groups these five attributes because modern robotic systems must simultaneously satisfy multiple requirements: intelligence for autonomous decision-making, adaptability for dynamic environments, security for data protection, ethical compliance for societal acceptance, and resilience for operational reliability. They are interconnected—weakness in one area can compromise the entire system—though their relative importance may vary by application.

10.
The most significant paradigm shift is from viewing robots as isolated tools to understanding them as interconnected nodes within complex Cyber-Physical Systems. This is evidenced in paragraph one (the holistic perspective) and paragraph two (CPS framework), where the focus shifts from individual performance optimization to interoperability, uncertainty management, and ethical integration within larger ecosystems.

Part B: Fill in the Gaps — Answer Key

Item	Answer
1	deterministic
2	interoperability
3	probabilistic
4	fault-tolerant
5	graceful degradation
6	feedback loops
7	monolithic
8	separation of concerns
9	sensor fusion
10	modularity